SAI GANESH VUPPALA

AI Architect | AWS Solutions Architect | Amazon Connect Architect & Developer | DevOps Engineer

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Certified: CEH (EC-Council) • CCNA (Cisco) • Google Cloud Professional Data Engineer

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PROFESSIONAL SUMMARY

AI and AWS Solutions Architect with 7+ years delivering cloud, contact center, and DevOps platforms across Healthcare, Financial Services, Energy, and Retail. Architects Generative AI on Amazon Bedrock (Knowledge Bases, Agents, Guardrails, RAG) and embeds it in Amazon Connect chat and voice. Leads enterprise migrations from Genesys and LivePerson to Amazon Connect through hybrid coexistence, number porting, and production cutover. Owns infrastructure end to end with reusable Terraform modules and GitHub Actions CI/CD, secured for HIPAA and PCI-DSS.

AREAS OF EXPERTISE

- **AI Architect:** Bedrock Knowledge Bases, Agents with Lambda action groups, Guardrails, RAG pipelines, evaluation harnesses.
- **AWS Solutions Architect:** Serverless, event-driven design on Lambda, Step Functions, API Gateway, EventBridge, Kinesis, DynamoDB.
- **Amazon Connect Architect & Developer:** Target-state design, contact flows, omnichannel routing, telephony, Agent Workspace apps.
- **DevOps Engineer:** Terraform modules, remote state, drift detection, GitHub Actions with OIDC and approval-gated promotion.

TECHNICAL SKILLS

Generative AI & RAG: Amazon Bedrock (Knowledge Bases, Agents, Action Groups, Guardrails, Prompt Engineering), RAG (chunking, Titan Embeddings, semantic retrieval, re-ranking), OpenSearch Serverless, Kendra, Amazon Q in Connect, Lex V2, Transcribe, Comprehend, Polly

Amazon Connect: Contact Flows & Modules, Omnichannel Voice/Chat/Task Routing, IVR, Queues, Routing Profiles, Telephony (DID/Toll-Free Porting, Carrier/SIP Routing), Agent Workspace, CCP & Streams API, Contact Lens, Customer Profiles, CTRs, Metrics APIs, Genesys and LivePerson Migration, Hybrid Coexistence & Cutover

DevOps, IaC & CI/CD: Terraform (modules, remote state, workspaces, drift detection), GitHub Actions (OIDC, sandbox → dev → stage → prod, approval gates), CloudFormation, Jenkins, CodePipeline, CodeBuild, SonarQube, Docker, Kubernetes, OpenShift, Git

AWS Platform: Lambda, Step Functions, API Gateway, EventBridge, SQS, SNS, Kinesis Data Streams & Firehose, ECS & Fargate, EKS, EC2, DynamoDB, S3, Athena, Glue, OpenSearch, QuickSight, Redshift, VPC, Route 53, CloudFront, Transit Gateway, Direct Connect

Security & Compliance: IAM (roles, permission boundaries, SCPs), KMS, Secrets Manager, VPC Endpoints, PrivateLink, WAF, HIPAA, PCI-DSS, RBAC, Security Assessment & Hardening (CEH)

Observability: CloudWatch (Alarms, Logs Insights, Dashboards), X-Ray, CloudTrail, ELK, DataDog, New Relic, Load & Regression Testing, Incident Management & RCA

Languages & Platforms: Python (primary), Node.js, Java (Spring Boot), Bash, React; Linux (Amazon Linux, Ubuntu, RHEL), Windows Server; GCP (Professional Data Engineer), Azure

PROFESSIONAL EXPERIENCE

Amazon Connect Architect (GenAI) | Miratech | UnitedHealth Group

January 2026 – Present

- Architect the enterprise migration of chat and voice from LivePerson and Genesys to Amazon Connect across multiple lines of business, owning target-state design, routing model, and integration blueprint.
- Replaced legacy FAQ bots with a Bedrock RAG assistant: policy and benefits content on S3, embedded with Titan, indexed in OpenSearch Serverless, and served as citation-backed answers in Connect chat via a Knowledge Base.
- Develop Bedrock Agents with Lambda action groups calling member APIs for eligibility, claim status, and case lookup, with Guardrails enforcing PHI-safe responses and evaluation harnesses tuning relevance and containment.
- Design Genesys–Connect hybrid coexistence for phased cutovers: carrier routing splits, cross-platform transfers with context passing, DID/toll-free porting plans, and rollback runbooks.
- Author reusable Terraform modules for Connect instances, contact flows, queues, routing profiles, Lambda/Lex associations, Kinesis streams, and IAM roles, with remote state and drift detection.
- Own delivery through GitHub Actions with OIDC role assumption, running fmt/validate/plan on pull requests and promoting applies from sandbox through production behind approval gates.
- Built a "ghost agent" auto-offline fix (agent event stream, Kinesis, Lambda, DynamoDB, EventBridge) that forces stale CCP sessions offline to protect routing accuracy.
- Designed a Repeat Contact Rate pipeline (CTR stream, Kinesis, Lambda, DynamoDB) surfaced in a custom Contact Intelligence app embedded in the Connect Agent Workspace.
- Enforce HIPAA-aligned security with IAM permission boundaries, SCPs, KMS, Secrets Manager, and VPC isolation; run integration, regression, and load testing before each cutover.
- Partner with LOB stakeholders, client architects, and AWS specialists on design reviews, and mentor engineers on Connect flow patterns, Bedrock/RAG, and Terraform delivery.

Amazon Connect Developer | Enbridge Gas *May 2024 – December 2025*

- Developed Connect contact flows for customer service, outage reporting, and billing, with IVR menus, natural-language capture, priority routing, callbacks, and emergency call treatment.
- Integrated Bedrock Knowledge Bases into agent-assist with a RAG ingestion pipeline (S3, embeddings, OpenSearch vector index refreshed by EventBridge-triggered Lambda) serving grounded answers during live contacts.
- Deployed all Connect infrastructure with Terraform and GitHub Actions (lint, plan, apply on merge, per-environment secrets), eliminating console drift and cutting release cycles from days to hours.
- Built Lex V2 bots with Lambda fulfillment against billing and account systems, and standardized error handling so flows degraded gracefully instead of dropping contacts.
- Streamed Contact Trace Records through Kinesis and Firehose into S3, cataloged with Glue and queried with Athena for contact volume, handle time, and abandonment reporting in QuickSight.
- Enabled Contact Lens sentiment, category rules, and redaction; applied least-privilege IAM, KMS, and VPC endpoints.

Amazon Connect Engineer | Capital One, TX *May 2023 – May 2024*

- Supported the Genesys-to-Amazon Connect transition for card servicing, rebuilding IVR flows, skills-based routing, and queue logic in Connect and validating parity before each cutover.
- Executed telephony changes for migration waves (DID/toll-free porting, carrier routing updates, Genesys–Connect transfers) and ran production cutovers with rollback runbooks.
- Implemented in-flow customer verification with Lambda lookups against core banking APIs through API Gateway, passing verified context to agents.
- Integrated Lex bots and early Bedrock prototypes into chat for intent triage and wrap-up summaries, plus a retrieval pipeline over policy documents for agent answer suggestions.
- Delivered infrastructure with Terraform and CloudFormation and CI/CD with GitHub Actions, CodePipeline, and CodeBuild for zero-downtime releases with automated rollback.
- Enforced PCI-DSS controls with IAM role boundaries, KMS encryption, redacted logging, and network segregation; led incident response and root cause analysis during high-volume periods.

AWS Engineer | Accenture, India *January 2021 – December 2022*

- Migrated legacy systems to cloud-native, API-driven AWS backends in Python and Java (Spring Boot), with event-driven pipelines on Lambda, Step Functions, EventBridge, and Kinesis.
- Created reusable Terraform and CloudFormation modules and CI/CD pipelines (GitHub Actions, Jenkins, CodePipeline) with automated deployment and rollback.
- Containerized microservices with Docker and Amazon EKS, with observability through CloudWatch, X-Ray, CloudTrail, and ELK.

DevOps Engineer | Morris Garages *June 2018 – December 2020*

- Built CI/CD pipelines with Jenkins and GitHub Actions automating build, test, and deployment across staging and production.
- Authored CloudFormation templates for infrastructure provisioning, reducing deployment time and eliminating configuration drift.
- Containerized applications with Docker on Kubernetes (EKS) and built serverless services with Lambda, API Gateway, SQS, SNS, and EventBridge.

CERTIFICATIONS & EDUCATION

- **Certified Ethical Hacker (CEH)** – EC-Council • **Cisco Certified Network Associate (CCNA)** – Cisco • **Professional Data Engineer** – Google Cloud
- **Master's in Computer Science** – Central Michigan University, Mount Pleasant, MI